

Gastrointestinal Imaging

Mannudeep K. Kalra, MD,
DNB
Michael M. Maher, MD, FFR
(RCSI), FRCR
Thomas L. Toth, DSc
Ravi S. Kamath, MD, PhD
Elkan F. Halpern, PhD
Sanjay Saini, MD

Index terms:

Abdomen, CT, 78.1211
Computed tomography (CT),
radiation exposure
Computed tomography (CT),
technology
Computed tomography (CT), thin-
section, 78.12118, 88.12118
Pelvis, CT, 88.1211

Published online

10.1148/radiol.2322031304
Radiology 2004; 232:347–353

¹ From the Division of Abdominal Imaging and Intervention, Department of Radiology, Massachusetts General Hospital and Harvard Medical School, White 270-E, 55 Fruit St, Boston, MA 02114 (M.K.K., M.M.M., R.S.K., E.F.H., S.S.); and GE Medical Systems, Waukesha, Wis (T.L.T.). From the 2003 RSNA scientific assembly. Received August 15, 2003; revision requested October 13; revision received October 30; accepted January 5, 2004. Supported in part by a grant from GE Medical Systems, Waukesha, Wis. Address correspondence to M.K.K. (e-mail: mannudeep_k_kalra@yahoo.com).

Author contributions:

Guarantor of integrity of entire study, S.S.; study concepts and design, M.K.K., S.S.; literature research, M.K.K.; clinical studies, M.K.K., M.M.M.; experimental studies, T.L.T.; data acquisition, M.K.K., R.S.K.; data analysis/interpretation, M.K.K., M.M.M., E.F.H.; statistical analysis, M.K.K., M.M.M., E.F.H.; manuscript preparation and definition of intellectual content, M.K.K.; manuscript editing, M.K.K., S.S., T.L.T., M.M.M.; manuscript revision/review, M.K.K., M.M.M.; manuscript final version approval, all authors

© RSNA, 2004

Comparison of Z-Axis Automatic Tube Current Modulation Technique with Fixed Tube Current CT Scanning of Abdomen and Pelvis¹

PURPOSE: To compare image quality, diagnostic acceptability, and radiation exposure associated with 16-section multi-detector row computed tomographic (CT) examinations of abdomen and pelvis performed with z-axis modulation technique of automatic tube current modulation and with manual selection of fixed tube current.

MATERIALS AND METHODS: Sixty-two consecutive subjects (mean age, 60 years; age range, 19–84 years; male-to-female ratio, 35:27) underwent follow-up CT of abdomen and pelvis with use of a 16-section multi-detector row scanner and z-axis modulation technique (10.5–12.0-HU noise index, 10–380 mA). Scanning parameters included 140 kVp, 0.5–1.0-second gantry rotation time, 0.938:1 beam pitch, and 5-mm reconstructed section thickness. For each subject, images obtained with z-axis modulation were compared with previous images obtained with fixed tube current (200–300 mA) and with other parameters identical. Images were compared for noise and diagnostic acceptability by two subspecialty radiologists using a five-point scale (1, unacceptable; 3, acceptable; 5, excellent) at five levels: upper liver at diaphragm, porta hepatis, right kidney hilum, iliac crest, and upper margin of acetabulum. Tube current and gantry rotation time used for acquisitions at these levels were recorded. Data were analyzed with parametric and nonparametric statistical tests.

RESULTS: Although no significant differences were found ($P = .34$), images acquired with z-axis modulation at the levels of the upper liver (diaphragm) and acetabulum had a higher noise and lower diagnostic quality, compared with images acquired with fixed tube current. Compared with fixed tube current, z-axis modulation resulted in tube current–time product reduction in 54 (87%) of 62 examinations (mean reduction, 71.2 mAs) and increase in eight (13%) (mean increase, 17.0 mAs).

CONCLUSION: Compared with manually selected fixed tube current, z-axis automatic tube current modulation resulted in reduced tube current–time product and similar image noise and diagnostic acceptability at CT of abdomen and pelvis.

© RSNA, 2004

Computed tomographic (CT) scanning techniques must aim for optimum image quality at the lowest possible radiation exposure to patients (1). However, the rapid evolution of CT technology, with the development of single-detector row spiral CT scanners and, subsequently, multi-detector row CT scanners, has made the selection of appropriate scanning parameters an arduous task (2,3). In particular, with changes in hardware and in available scanning parameters, manual selection of tube current (an important determinant of

image quality as well as radiation exposure) for specific study protocols in subjects who have different body habitus has become difficult.

Automatic tube current modulation techniques represent technologic development, which aims to facilitate the optimization of image quality and radiation exposure, thus eliminating arbitrary selection of tube current by CT technologists (2). The purpose of this study was to compare image quality, diagnostic acceptability, and radiation exposure associated with 16-section multi-detector row CT examinations of abdomen and pelvis performed with the z-axis modulation technique of automatic tube current modulation and with manual selection of fixed tube current.

MATERIALS AND METHODS

Study Subjects and CT Parameters

The human research committee of the institutional review board approved the study protocol with a waiver of informed consent. Sixty-two consecutive subjects underwent follow-up contrast material-enhanced CT examinations of the abdomen and pelvis with a multi-detector row CT scanner (LightSpeed 16; GE Medical Systems, Waukesha, Wis) with 16 channels and manufacturer-supplied software (M3; GE Medical Systems). CT was performed by using z-axis modulation (Auto mA; GE Medical Systems). All follow-up CT studies were clinically indicated for assessment of abdominal and pelvic lesions. A total of 35 men (mean age, 56 years; age range, 26–84 years) and 27 women (mean age, 50 years; range, 19–84 years) constituted the study cohort. The mean age of subjects in the study cohort was 53 years (range, 19–84 years). Weight of subjects in the study cohort was recorded prior to CT. Mean weight of the subjects in the study cohort was 78.9 kg (mean weight of male subjects, 80.9 kg; mean weight of female subjects, 69.9 kg), with a range of 45.4–131.5 kg.

The standard departmental protocol was used for scanning with z-axis modulation. Parameters for CT examinations performed with z-axis modulation included 140 kVp, 0.938:1 beam pitch, 16×1.25 -mm detector configuration, 18.75-mm table feed per gantry rotation, 5-mm reconstructed section thickness, and standard reconstruction algorithm. When z-axis modulation is used, a noise index and maximum and minimum tube current thresholds must be selected. In

this study, noise indices of 10.5–12.0 HU, which were recommended by the manufacturer for standard abdominal CT, were selected with the following distribution: 10.5 HU was selected in five cases, 11.0 HU was selected in 20 cases, 11.5 HU was selected in 32 cases, and 12.0 HU was selected in five cases. The minimum and maximum tube current thresholds were 10 and 380 mA, respectively. With z-axis modulation, previewing of tube current values that will be used for scanning for selected noise index prior to actual scanning can help to prevent saturation at the maximum tube current threshold (380 mA at 0.5-second gantry rotation time was selected for the standard protocol). Such saturation may occur in subjects with a large body habitus, for whom a greater tube current–time product (380 mA $\times$ 0.5 sec = 190 mAs) may be needed to achieve the requested noise index, or when a low noise index (high image quality) is desired. Therefore, to maintain image quality at the desired noise index by increasing the tube current–time product, technologists were instructed to increase rotation time until saturation was prevented ($n = 4$). Subjects were scanned with a 0.5-second gantry rotation time, which was increased to 1 second to prevent saturation at the maximum tube current threshold (to enable achievement of the desired noise index at the required tube current–time product instead of limiting the tube current–time product to the maximum of 190 mAs at 0.5-second rotation time).

Comparisons

For each subject, CT images obtained with the z-axis modulation technique were compared with images from previous CT examinations with a 16-section multi-detector row scanner performed according to our standard departmental protocol for CT, which includes manual selection of fixed tube current (200–300 mA at 0.8–1.0-second tube rotation time). The remaining parameters for examinations performed with manual selection of fixed tube current were identical to those for CT with the z-axis modulation technique. The mean interval between the two examinations was 5 months (range, 2–8 months).

Image Analysis

For each subject, a direct side-by-side comparison of images acquired with the z-axis modulation technique and with manual selection of fixed tube current

was performed at a digital picture-archiving system diagnostic workstation (Impax RS-3000 1K; Agfa, Ridgefield Park, NJ) by two subspecialty radiologists blinded to the scanning techniques used (M.M.M., with 5 years of experience; M.K.K., with 3 years of experience). Each radiologist independently graded the studies for image noise and diagnostic acceptability by using a five-point scale (1, unacceptable; 2, minimally acceptable; 3, acceptable; 4, highly acceptable; 5, excellent) to score images at five anatomic levels, including the upper liver at the level of the diaphragm, porta hepatis, right kidney hilum, iliac crest, and upper margin of the acetabulum. Acceptable image noise was defined as an average amount of mottle or graininess, with satisfactory visualization of small anatomic structures such as blood vessels and interface between structures of variable attenuation. Unacceptable image noise was defined as mottle that interfered with visualization of these structures, whereas excellent image noise was defined on the basis of minimal or no appreciable mottle. Diagnostic quality was considered acceptable if sharpness of different structures, contrast resolution, and lesion visualization were satisfactory. Diagnostic quality was described as unacceptable if these image attributes were unsatisfactory, or as excellent if visualization was considerably superior. Images from all examinations were assessed at the same window level and window width (40 and 400 HU, respectively). The degree of graininess or mottle on the image was the main factor considered in the scoring of image noise. Diagnostic acceptability was graded on the basis of confidence in diagnosis of disease at that level, which usually depended on soft-tissue contrast, sharpness of tissue interfaces, lesion conspicuity, and degree of image degradation by streak or beam-hardening artifacts. In addition, both readers assessed images for the presence of abnormalities and artifacts.

Radiation Exposure

After the image quality assessment, tube currents and gantry rotation time used for acquiring images at each level (upper liver at the level of the diaphragm, porta hepatis, right kidney hilum, iliac crest, and upper margin of the acetabulum) were recorded (R.S.K.) for examinations performed with both techniques. The tube current–time product was calculated for each anatomic level and for

TABLE 1
Comparison of Image Noise and Diagnostic Acceptability Scores

Anatomic Level	Z-Axis Modulation		Fixed Tube Current	
	Image Noise	Diagnostic Acceptability	Image Noise	Diagnostic Acceptability
Upper liver	2.8 ± 0.7	3.0 ± 0.6	2.9 ± 0.7	3.1 ± 0.8
Porta hepatis	2.9 ± 0.6	3.2 ± 0.6	2.8 ± 0.7	3.2 ± 0.7
Renal hilum	2.8 ± 0.7	3.2 ± 0.7	2.7 ± 0.7	3.2 ± 0.7
Iliac crest	3.0 ± 0.8	3.3 ± 0.6	2.9 ± 0.7	3.4 ± 0.7
Acetabulum	2.9 ± 0.7	3.3 ± 0.6	3.0 ± 0.7	3.4 ± 0.5

Note.—Data are means and standard deviations. No statistically significant differences were found between CT images acquired with z-axis automatic modulation and those acquired with fixed tube current ($P = .34-.84$).

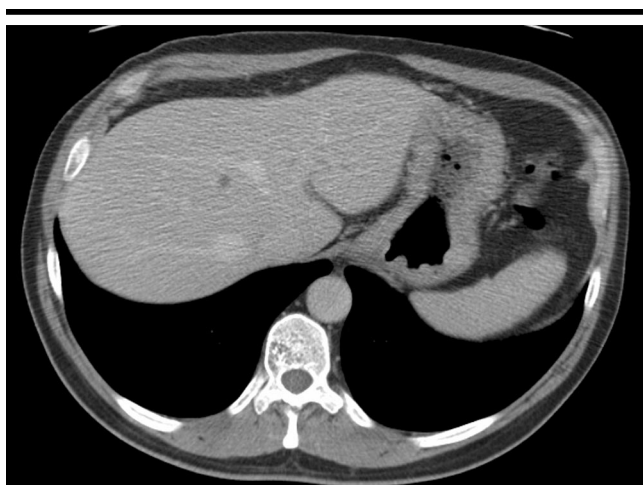


Figure 1. Transverse CT image acquired with z-axis modulation at 52 mAs in a 79-year-old man (weight, 78 kg) with prostate cancer shows noise (streak artifacts) in the upper part of the liver, at the level of the diaphragm.

each examination by multiplying the tube current by the gantry rotation time.

Statistical Analysis

Image quality scores for noise and diagnostic acceptability with z-axis tube current modulation versus manual selection of fixed tube current were compared by using the Wilcoxon signed rank test and statistical software (SAS/STAT; SAS Institute, Cary, NC). Tube current–time products for each anatomic level and for examinations performed with z-axis modulation versus manual selection of fixed tube current were compared by using a paired t test (Excel; Microsoft, Redmond, Wash). Differences in mean tube current–time product between examinations performed with z-axis modulation and those performed with manual selection of fixed tube current techniques were also calculated as means and ranges by using the same software program. Linear statistical correlation between tube

current used with z-axis modulation (or fixed tube current) and weight of subjects was also performed. In addition, the mean tube current–time products with fixed tube current and with z-axis modulation were calculated separately for male and female subjects. A Student t test was used to test for differences in subjects' body weight and in mean tube current–time product with the two scanning techniques according to sex. The degree of interobserver concordance was determined with calculation of κ statistics. A P value of less than .05 was considered to indicate a statistically significant difference.

RESULTS

Image Noise and Diagnostic Acceptability

Image noise and diagnostic acceptability scores (mean ± standard deviation) for images obtained with z-axis modula-

tion and with manual selection of fixed tube current at the five anatomic levels are summarized in Table 1. At multi-detector row CT in the upper liver at the level of the diaphragm and at the acetabulum with z-axis modulation, image noise was slightly greater and diagnostic acceptability was slightly inferior to that with manual selection of fixed tube current, but the differences were not significant ($P = .34$ for both). At the levels of the porta hepatis, right kidney hilum, and iliac crest, no statistically significant difference was found in scores for image noise and diagnostic acceptability between images obtained with z-axis modulation and those obtained with fixed tube current ($P = .84$). At the level of the upper liver, however, both readers reported an increased frequency of streak artifacts with z-axis modulation (Fig 1). Eighty-one lesions were detected in the study cohort, which included liver lesions ($n = 19$), renal calculi ($n = 11$), kidney cysts ($n = 10$), adrenal lesions ($n = 9$), uterine adnexal masses ($n = 9$), bowel lesions ($n = 7$), gallbladder stones ($n = 7$), retroperitoneal lymphadenopathy ($n = 5$), psoas muscle masses ($n = 2$), and splenic calcifications ($n = 2$). All lesions detectable on CT images obtained with manual selection of fixed tube current were also seen on images obtained with z-axis modulation.

Comparisons

An overall reduction of 31.9% (range, –18.8% to 87.5%) in mean tube current–time product (59.8 mAs) was noted with z-axis modulation, compared with fixed tube current (Figs 2–4). The data for tube current–time product (mean ± standard deviation) with z-axis modulation and with manual selection of fixed tube current at the level of the upper liver, porta hepatis, right kidney hilum, iliac crest, and superior margin of the acetabulum are summarized in Tables 2 and 3. A stronger statistical correlation was found between patient weight and tube current with z-axis modulation (correlation coefficient, 0.75; $P < .001$) than with fixed tube current (correlation coefficient, 0.28; $P = .026$). Compared with the manual selection of fixed tube current, the use of the z-axis modulation technique resulted in a reduction in tube current–time product in 54 (87%) of 62 examinations, with a mean reduction of 71.2 mAs (2.0–168.1 mAs), which was equivalent to a 37.9% reduction in mean tube current–time product (Table 2). In the subset of examinations in which a reduction in

tube current–time product occurred with z-axis modulation, the tube current–time product range for z-axis modulation was 20–228 mAs, compared with 100–240 mAs for the fixed tube current technique. Overall, mean tube current–time product values with and without z-axis modulation were 116.71 and 187.90 mAs, respectively ($P < .001$) (Fig 5). In this group of examinations, the ranges for tube current–time product with the z-axis modulation technique at the five anatomic levels were 21–228 mAs (upper liver), 22.5–228 mAs (porta hepatis), 20–228 mAs (kidney hilum), 20.5–228 mAs (iliac crest), and 32–228 mAs (acetabulum). Mean and median patient weights for examinations in which a reduction in tube current–time product occurred with the z-axis modulation technique were 77.6 and 77.5 kg, respectively.

An increase in tube current–time product with z-axis modulation was noted in eight (13%) of 62 examinations, with a mean increase of 17.0 mAs (1.6–36.0 mAs), which was equivalent to an 11.6% increase over tube current–time product with manual selection of fixed tube current (Table 3). The tube current–time product ranges with the z-axis modulation technique in this subset of examinations at different levels were 104–228 mAs (upper liver), 124–228 mAs (porta hepatis), 138–228 mAs (kidney hilum), 148.5–228 mAs (iliac crest), and 122.5–228 mAs (acetabulum). Mean scores for image noise with z-axis modulation and fixed tube current techniques in these eight examinations were 3.1 and 3.2, and mean scores for diagnostic acceptability were 2.3 and 2.8, respectively ($P = .01$). The mean and median weights of patients in this category were 90.3 and 91.2 kg—significantly greater than those of patients for whom tube current–time product was reduced with z-axis modulation ($P = .034$) (Fig 6).

For male patients the mean reduction in tube current–time product with z-axis modulation was 35.8 mAs (mean tube current–time products with fixed tube current and z-axis modulation were 188.1 and 152.3 mAs, respectively), whereas for female patients the mean reduction with z-axis modulation was 50.2 mAs (mean tube current–time products with fixed tube current and z-axis modulation were 179.5 and 129.3 mAs, respectively). Although the reductions in tube current–time product with z-axis modulation for male and female patients were not significantly different ($P = .11$), the mean tube current–time

product with z-axis modulation for female patients was significantly lower than that for male patients ($P = .01$). With fixed tube current, no statistically significant difference was found in

mean tube current–time product between male and female patients ($P = .2$), despite a significant difference in weight between the two groups ($P = .01$).

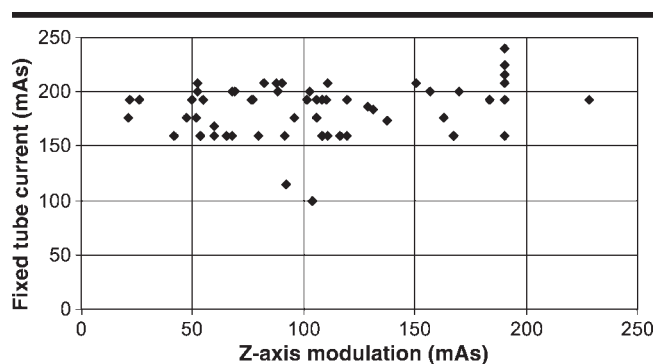


Figure 2. Scatterplot of tube current–time products for examinations performed at the level of the upper liver shows that, in the majority of examinations, z-axis modulation resulted in reduced tube current–time product compared with that for manual selection of fixed tube current.

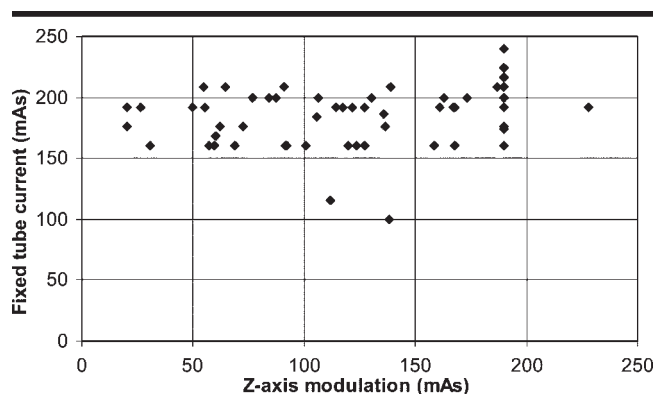


Figure 3. Scatterplot of tube current–time products for examinations performed at the level of the renal hilum shows that, in the majority of examinations, z-axis modulation resulted in reduced tube current–time product compared with that for manual selection of fixed tube current.

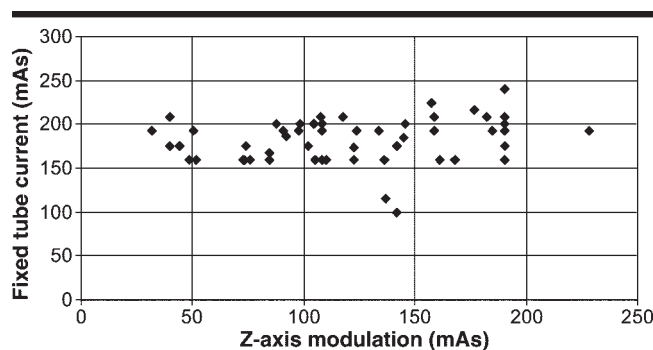


Figure 4. Scatterplot of tube current–time products for examinations performed at the level of the acetabulum shows that, in the majority of examinations, z-axis modulation resulted in reduced tube current–time product compared with that for manual selection of fixed tube current.

TABLE 2
Comparison of Reductions in Tube Current–Time Product with Z-Axis Automatic Modulation versus Fixed Tube Current

Anatomic Level	Tube Current–Time Product with Z-Axis Modulation	Tube Current–Time Product with Fixed Tube Current	Tube Current–Time Product Reduction with Z-Axis Modulation*
Upper liver	103.6 ± 50.3	187.9 ± 21.4	84.3 ± 45.7 (44.9)
Porta hepatis	121.0 ± 51.6	187.9 ± 21.4	66.9 ± 47.7 (35.6)
Renal hilum	119.9 ± 54.2	187.9 ± 21.4	68.0 ± 49.4 (36.2)
Iliac crest	115.4 ± 56.4	187.9 ± 21.4	72.5 ± 51.7 (38.6)
Acetabulum	123.8 ± 50.0	187.9 ± 21.4	64.1 ± 47.1 (34.1)

Note.—Data are mean milliamperere-seconds ± standard deviations unless otherwise noted.

* Data in parentheses indicate percent change. For examinations in which mean tube current–time product decreased with z-axis modulation compared with fixed tube current, the difference was statistically significant ($P < .001$).

TABLE 3
Comparison of Increases in Tube Current–Time Product with Z-Axis Automatic Modulation versus Fixed Tube Current

Anatomic Level	Tube Current–Time Product with Z-Axis Modulation	Tube Current–Time Product with Fixed Tube Current	Tube Current–Time Product Reduction with Z-Axis Modulation*
Upper liver	165.1 ± 41.2	158.5 ± 26.4	6.6 ± 27.9 (4.0)
Porta hepatis	179.7 ± 30.4	158.5 ± 26.4	6.2 ± 13.2 (11.8)
Renal hilum	181.5 ± 26.9	158.5 ± 26.4	23.0 ± 14.5 (12.7)
Iliac crest	180.4 ± 24.9	158.5 ± 26.4	21.9 ± 16.9 (12.2)
Acetabulum	170.8 ± 35.4	158.5 ± 26.4	12.3 ± 29.0 (7.2)

Note.—Data are mean milliamperere-seconds ± standard deviations unless otherwise noted.

* Data in parentheses indicate percent change. For examinations in which mean tube current–time product increased with z-axis modulation compared with fixed tube current, the difference was not statistically significant ($P = .31$).

Interobserver Agreement

Substantial interobserver agreement was noted between the two radiologists for assessment of both image noise and diagnostic acceptability (weighted κ coefficient, 0.7; $P < .05$).

DISCUSSION

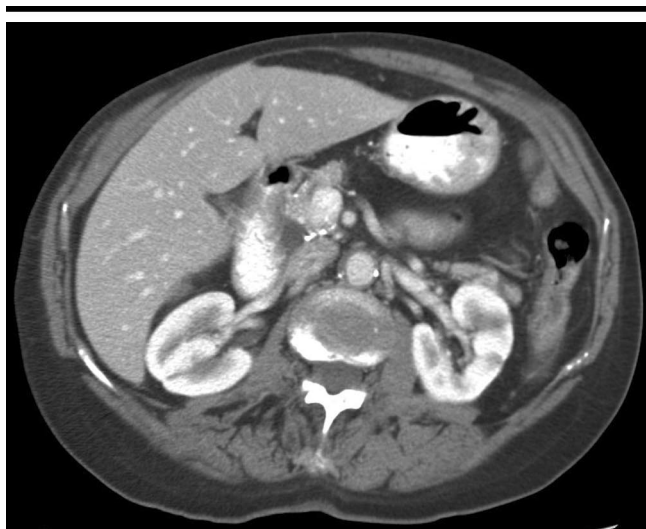
Automatic tube current modulation was developed to minimize subjective estimation and selection of tube current required to obtain desired image quality at reasonable radiation exposure levels (2–5). With the real-time angular modulation technique (CARE Dose; Siemens Medical Solutions, Forchheim, Germany), the direction of the tube current is adjusted with regard to an individual section to minimize x-rays in projections that cause less beam attenuation, and, thus, a lower tube current is needed to maintain constant image noise (6–9). In one reported study of 100 pediatric CT examinations, a 10%–60% decrease in effective tube current–time product was found with this technique, with a mean

reduction of 22.3% for all examinations (neck, 20%; thorax, 23%; abdomen, 23%; thorax and abdomen, 22%), without a loss of image quality (10). With z-axis modulation—the most recently developed technique for automatic tube current modulation—the tube current is adjusted so that all images will have a similar user-selected quantum noise level (noise index) independent of patient size and anatomy (2). With this technique, users select the amount of noise that will be present in the reconstructed images. The noise index value is approximately equal to the image noise (standard deviation) in the central region of the image obtained from CT of a uniform phantom. The tube current modulation for each anatomic region is determined with calculations of patient density, size, and shape (beam attenuation) performed by using a single scout projection acquired before the actual examination and a set of empirically determined noise prediction coefficients for the reference CT technique. The reference scanning technique comprises an arbitrary 2.5-mm section thickness at the selected tube potential and

200 mAs at transverse reconstruction with a standard reconstruction algorithm. With the reference noise and the difference between the reference and the selected techniques, the tube current required to obtain a prescribed noise index is calculated by using known physics equations that state that noise is inversely proportional to the square root of the number of photons and that the number of photons is proportional to the section thickness, section acquisition time, and tube current. The actual noise measured in a region of interest on an image often differs from the selected noise index because noise index settings only adjust the tube current, whereas the image noise also is affected by parameters such as the reconstruction algorithm, reconstructed section thickness (if different from prospective section thickness), variations in patient anatomy, and beam-hardening artifacts.

To the best of our knowledge, in no previous peer-reviewed scientific study has there been a comparison of z-axis modulation with fixed tube current scanning. Findings in the present study show that the z-axis modulation technique provides images with satisfactory image noise and diagnostic acceptability for CT of the abdomen and pelvis, compared with those achieved with the fixed tube current technique. These findings are in agreement with those reported for studies with the angular modulation technique (6–10). We noted with interest that image noise and diagnostic acceptability of images obtained with z-axis modulation at the level of the upper liver were slightly inferior to those of images obtained with the fixed tube current technique, although the difference was not significant statistically. This finding can probably be explained by the fact that there is a marked decrease in tube current at the transition point between the low-attenuation lung parenchyma and higher-attenuation subdiaphragmatic organs.

From a radiation dose reduction perspective, in comparison with the fixed tube current technique, z-axis modulation showed a 31.9% (–18.8% to 87.5%) decrease in the mean tube current–time product. A 34.1%–44.9% reduction in mean tube current–time product was observed with the z-axis modulation technique, compared with the manual selection of tube current. Comparatively lower decreases in tube current–time product at the level of the acetabulum may be explained on the basis of higher attenuation caused by the higher tube current–time product needed to generate comparable quantum



a.



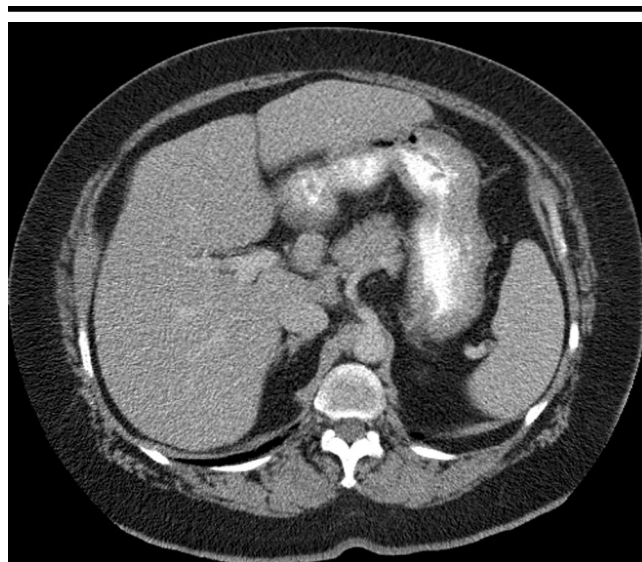
b.

Figure 5. Transverse CT images at the level of the renal hilum in a 60-year-old woman (weight, 70 kg) with chronic pancreatitis, acquired with (a) fixed tube current at 208 mAs and (b) z-axis modulation at 112 mAs. Both images were rated acceptable for image noise and diagnostic quality.

noise on images of pelvic bone. Results of previous studies (10,11) have shown a mean reduction of 20%–23% in radiation exposure for abdominal and pelvic CT examinations performed with the angular modulation technique, compared with those performed with the fixed tube current technique. Our study results show a reduction in tube current–time product with z-axis modulation that is comparable with that achieved with the angular modulation technique. Because the reference fixed tube current CT examinations, patient cohorts, and scanner types used in the studies evaluating the two techniques were different, this comparison may not

indicate an actual difference in radiation savings. Our study results, however, suggest the potential value of combining the two complementary techniques of automatic tube current modulation to benefit from the radiation savings of both techniques (12,13).

In examinations in which the use of z-axis modulation resulted in increased radiation exposure compared with radiation exposure with manual selection of fixed tube current, a 4.0%–12.7% increase was noted in mean tube current–time product with z-axis modulation. Pa-



a.



b.

Figure 6. Transverse CT images obtained for colorectal cancer staging in a 74-year-old man (weight, 115 kg). (a) Image acquired with fixed tube current at 192 mAs was rated minimally acceptable for image noise and diagnostic quality. (b) Corresponding image acquired at 228 mAs with z-axis modulation was scored as acceptable for image noise and diagnostic quality.

tients in these examinations, however, were much heavier ($P = .03$) than patients in the examinations with z-axis modulation that resulted in tube current–time product reduction. In the former subset of examinations, a significant improvement ($P = .01$) in image noise and diagnostic acceptability with z-axis modulation technique may have been due to increased tube current–time product with z-axis modulation or to poorer image quality with manual selection of inappropriately low tube current–time product. It is interesting to note that

a below-average quality score (ie, <3) for fixed tube current technique in these examinations makes the latter possibility seem more likely. This finding demonstrates an advantage with the use of z-axis tube current modulation for maintenance of the selected noise index in larger patients. Although technologists can set the maximum tube current threshold for z-axis modulation at the same level as for fixed tube current protocols to prevent unexpectedly high tube currents, this can lead to increased noise at capped regions. To avoid surprises, especially when familiarizing themselves with the z-axis modulation technique, technologists can also preview the "mA table" at the console monitor prior to scanning.

The present study highlights some important considerations regarding the z-axis modulation technique. Although both the minimum tube current threshold applied to prevent an excessive drop in tube current at certain locations and the maximum tube current cap applied to prevent an excessive increase in tube current act as safety features, they may represent a departure from the goals of automation of radiation dose and optimization of image quality. In applying these limits, it is imperative that technologists preview the tube current selected by the technique and set arbitrary tube current thresholds dependent on the type of examination and the patient habitus. Thus, future variants of the techniques should be aimed at minimizing the subjective selection of tube current threshold to reduce noise and streak artifacts at certain levels. Although our study results show a greater radiation dose reduction with z-axis modulation compared with those reported for angular modulation, only a direct comparison of the two techniques can prove the benefit of one technique over the other. Currently, an important limitation of most available automatic tube current modulation techniques is that they do not offer a combined technique that incorporates both angular and z-axis modulation.

It is hoped that experience with z-axis modulation will result in more accurate selection of upper and lower tube current thresholds. Although our study results show that z-axis modulation results in a radiation dose reduction comparable with that achieved with angular modulation, this observation is not important, because the two techniques are complementary. Thus, manufacturers should be encouraged to deliver systems that incorporate the radiation dose reduction benefits of both z-axis modulation and angu-

lar modulation techniques. As these new techniques of tube current modulation are being introduced, it is important for radiologists and physicists to understand how image quality is affected by dynamic changes in tube current so that optimization of image quality at the lowest possible radiation dose can be satisfactorily realized.

There are some limitations in the present study. We evaluated examinations performed with z-axis modulation by using the noise index values recommended by the vendor and did not undertake an independent determination of the stated parameter. Although our study demonstrated a reduction in tube current-time product in CT examinations performed with z-axis modulation, a similar study with a larger study cohort and greater range of noise index values will be necessary to validate our findings. The slightly higher image noise and slightly inferior diagnostic acceptability at the level of the upper liver with z-axis modulation compared with fixed tube current scanning, although not significant statistically, need further investigation to assess whether the abrupt change in beam attenuation was the source of increased noise. The evaluation of five levels in the abdomen and pelvis was based on the premise that significant differences in the regional dimensions and beam attenuation were most likely to be observed in these regions. Although the significant variation in tube current between the anatomic levels assessed in our study represented the approximate range of variation in tube current with z-axis modulation, actual reduction or increment in radiation dose with respect to fixed tube current may differ from projected radiation exposure changes associated with z-axis modulation. Our study did not include comparison of standard radiation dose indices, CT dose index, dose length product, or estimated effective dose with z-axis modulation and fixed tube current techniques. Because quantitative indicators of image quality (eg, objective image noise) are affected by differences in patient positioning in the gantry isocenter, field of view, and change in patient habitus during follow-up, we did not compare quantitative indicators, although images were acquired with similar scanning parameters.

In summary, compared with the fixed tube current technique, z-axis modulation provided similar levels of image noise, diagnostic acceptability, and lesion detection, as well as tube current

reduction, for CT of the abdomen and pelvis in the majority of our patients.

References

1. The ALARA (as low as reasonably achievable) concept in pediatric CT intelligent dose reduction. Multidisciplinary conference organized by the Society of Pediatric Radiology, August 18–19, 2001. *Pediatr Radiol* 2002; 32:217–313.
2. Kalra MK, Maher MM, Toth TL, et al. Strategies for CT radiation dose optimization. *Radiology* 2004; 230:619–628.
3. Kalra MK, Prasad S, Saini S, et al. Clinical comparison of standard-dose and 50% reduced-dose abdominal CT: effect on image quality. *AJR Am J Roentgenol* 2002; 179:1101–1106.
4. Kopka L, Funke M, Breiter N, Hermann KP, Vosschenrich R, Grabbe E. An anatomically adapted variation of the tube current in CT: studies on radiation dosage reduction and image quality. *Rofo Fortschr Geb Rontgenstr Neuen Bildgeb Verfahr* 1995; 163:383–387.
5. Giacomuzzi SM, Erckert B, Schopf T, et al. The smart-scan procedure of spiral computed tomography: a new method for dose reduction. *Rofo Fortschr Geb Rontgenstr Neuen Bildgeb Verfahr* 1996; 165: 10–16.
6. Greess H, Wolf H, Baum U, Kalender WA, Bautz W. Dosage reduction in computed tomography by anatomy-oriented attenuation-based tube-current modulation: the first clinical results. *Rofo Fortschr Geb Rontgenstr Neuen Bildgeb Verfahr* 1999; 170:246–250.
7. Greess H, Wolf H, Baum U, et al. Dose reduction in computed tomography by attenuation-based on-line modulation of tube current: evaluation of six anatomical regions. *Eur Radiol* 2000; 10:391–394.
8. Greess H, Baum U, Wolf H, et al. Dose reduction in spiral-CT: detection of pulmonary coin lesions with and without anatomically adjusted modulation of tube current. *Rofo Fortschr Geb Rontgenstr Neuen Bildgeb Verfahr* 2001; 173:466–470.
9. Mastora I, Remy-Jardin M, Suess C, Scherf C, Guillot JP, Remy J. Dose reduction in spiral CT angiography of thoracic outlet syndrome by anatomically adapted tube current modulation. *Eur Radiol* 2001; 11: 590–596.
10. Greess H, Nömayr A, Wolf H, et al. Dose reduction in CT examination of children by an attenuation-based on-line modulation of tube current (CARE Dose). *Eur Radiol* 2002; 12:1571–1576.
11. Tack D, De Maertelaer V, Gevenois PA. Dose reduction in multidetector CT using attenuation-based online tube current modulation. *AJR Am J Roentgenol* 2003; 181:331–334.
12. Greess H, Wolf H, Suess C, Lutze J, Kalender WA, Bautz W. Automatic exposure control to reduce dose in subsecond multislice spiral CT: phantom measurements and clinical results (abstr). *Radiology* 2002; 225(P):593.
13. Horiuchi T. Study on 3D modulation auto mA (abstr). *Jpn Soc Radiol Technol* 2002; 78(P):166.